

Recommendations for Fixing of Step Performed and Sequence Diagram Based on Text Mining on SRS Digibrary

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Abstract— Only some applications that are built are successful once the client uses them. This problem occurs because there is an inconsistency between applications built with the client's needs, especially in the Software Requirements Specification (SRS) document as the reference data for software development. These losses can be avoided by implementing consistency in the contents of the SRS document. Therefore, before application development, the SRS document consistency process is crucial to match the client's needs and the developer's understanding of developing software. This study aims to propose recommendations for fixing artifacts based on the lowest value of consistency between Step Performed and Sequence Diagrams through a text-mining approach. The results of the validation and reliability tests using Gwet's AC1 formula through the Python programming language produced a Kappa Score of 0.500000, while through expert questionnaires of 0.159998. Based on the comparison of artifact consistency calculations, recommendations for fixing artifacts were made for the lowest consistency measurement values in the comparison of SP09 (d9) with SD09 (d23), which had a value of 0.840124.

Keywords—*Extraction, Step Performed, Use Case Description, Consistency, Sequence Diagram, Text Mining*

I. INTRODUCTION

Consistency between application development by developers and client's needs is the primary key in building an application so that the client will use the application. However, not all applications built are successful until they reach the stage of the client uses the application because there is an inconsistency between the applications built with the client's needs, especially in the SRS document as reference data for software development. These losses can be avoided by implementing consistency in the contents of the Software Requirements Specification (SRS) document. Therefore, before application development, the SRS document consistency process is a crucial stage to match the client's needs and the developer's understanding of software development.

Previously, there was research on measuring the similarity between Steps Performed, which describes the steps taken to accomplish the goal of the Use Case in Use Case Descriptions, and Sequence Diagrams in SRS documents [1]. The data used in research [1] consisted of five Step Performed documents from the Use Case Description and five Sequence Diagram documents. The similarity between the two artifacts was measured using Cosine Similarity and tested using Gwet's AC1. Based on this study, the research focuses on propose

recommendations for fixing artifacts SRS artifacts whose consistency wants to compare through text mining of objects in the SRS Digibrary document.

Digibrary is an application for borrowing and returning book collections at the online library. The SRS Digibrary document used in this study consists of 14 Steps Performed from the Use Case Description and 14 Sequence Diagrams. Then, the consistency of the document will be measured using Cosine Similarity and tested for validity and reliability using Gwet's AC1.

This study aims to propose recommendations for fixing artifacts based on the lowest value of consistency between Step Performed and Sequence Diagrams based on Unified Modeling Language (UML) rules. There are contributions to this research, namely:

- Conduct requirements elicitation and identification of Use Case Description and Sequence Diagram artifacts, then proceed with identifying these artifacts.
- Applying the text mining method to produce text preprocessing of artifacts in Extraction activities.
- Carry out the process of document weighting, measuring the value of consistency, and testing the results through a process of validity and reliability.
- Generate extraction values as a reference for consistency between the Step Performed and Sequence Diagrams.
- Fixing the Step Performed and Sequence Diagram was carried out based on the results of text-preprocessing which were still inconsistent.

II. RELATED WORK

A. Software Requirement Specification (SRS)

Software Requirement Specification (SRS) is a document regarding a collection of requirements and descriptions of a software system. A guide map providing direction to all software development parties is part of the SRS document [2, 3, 4]. SRS document creation needs to be done in the early stages of the software development and design process because this document contains all the requirements of the needs in developing a software system. Requirements for stakeholders' needs and the software system's design are determined in the SRS document so that the SRS provides significant benefits to the success of software development [5, 6]. The relationship between the system and the environment,

the system and practitioners, and system components from different perspectives is depicted as a diagram in the SRS document based on the Unified Modeling Language (UML) [7].

The Use Case Diagram is one of the UML diagrams in the SRS document. Use Case Diagrams present a graphical representation of the relationships of actors, objectives, and dependencies between Use Cases in the system. In general, Use Case Diagrams are used to display an overview of the system in building and developing applications because an overview of each Use Case contained in the system is visualized in this diagram [8]. Each Use Case from the Use Case Diagram can be transformed into a Use Case Description because the Use Case Description explains the process in more detail and indicates the possible benefits that actors (system users) can obtain when performing the services offered in the Use Case. Names of Use Cases, scenarios, actors, and flow of activities are components of Use Case Descriptions so that Use Case Descriptions can be presented in other forms [9, 10]. Another form of presenting a Use Case Description is a Sequence Diagram. The Sequence Diagram is one of the UML diagrams that represent the interaction between existing objects in the form of scenarios carried out in response to events in the system to produce certain outputs. Objects related to the operating process are usually arranged from left to right [11]. Actors, lifelines, activation boxes, objects, and messages are Sequence Diagram elements [12].

B. Text Extraction and Text Mining

Text extraction is a method of converting unstructured formatted text into structured text [13]. One example of the implementation of text extraction is the Step Performed text extraction from the Use Case Description in the SRS document. The role of this method is to obtain quality information from unstructured text documents [14]. The extracted data needs to be processed using the text mining method. Text mining can analyze research trends, such as information systems and technology management [15]. Asif et al. [3] proposed the extraction of related Functional Requirements processed through a text mining process. One of the data processing using the text mining method is text pre-processing. The text pre-processing process is carried out to find patterns and trends in data [16]. Meanwhile, text mining has four main stages: collecting data, text pre-processing, designing models iteratively through evaluation and validation processes, and forming appearances and interactions to verify modeling results [17, 18, 19].

C. Cosine Similarity

Cosine Similarity is a formula for calculating the similarity between two vectors based on the cosine angle. Then, the lowest value on the cosine is -1. The explanation of the Cosine Similarity formula for measuring between two vectors is as follows [20, 21, 22, 23].

$$\cos(a, b) = \frac{a \cdot b}{\|a\| \cdot \|b\|} = a \cdot \frac{a_1 b_1 + \dots + a_n b_n}{\sqrt{a_1^2 + \dots + a_n^2} \sqrt{b_1^2 + \dots + b_n^2}} \quad (1)$$

The two vectors in the Cosine Similarity formula are a and b . The formula is used to find the cosine correlation between two normalized vectors, namely normalized a, b ($\|a\| = 1, \|b\| = 1$). Apriyanto et al. [1] proposed similarity measurement using Cosine Similarity for five Sequence Diagrams extraction.

D. Gwet's AC1

Gwet's AC1 is a statistical calculation method based on measuring the agreement index between two experts to test the validity and reliability of the model formed through a questionnaire. The questionnaire needs to contain questions with the results of answers to information relating to research objectives and answering problems in research [24]. The steps for testing the validity and reliability of a model using Gwet's AC1 are as follows [1, 2]:

- Build a table containing the results of the requirements elicitation between two experts based on Table I.

TABLE I. STATEMENT OF ELICITATION RESULTS

Expert 1	Expert 2		
	Yes	No	Total
Yes	A	B	B1 = A+B
No	C	D	B2 = C+D
Total	A1 = A+C	A2 = B+D	N

- Calculate the observed agreement (P) based on Table II using the following formula.

$$P = \frac{A+D}{N} \quad (2)$$

- Calculate the chance-agreement probability or $e(\gamma)$ as follows.

$$e(\gamma) = 2P_1(1 - P_1) \quad (3)$$

with P_1 based on the formula as follows.

$$P_1 = \frac{(A_1+B_1)/2}{N} \quad (4)$$

Information:

N is the amount of data. Meanwhile, A_1 is the sum of Experts 1 and 2.

- Represent the results of possible expert estimates by grouping data into the "Yes" category.
- Calculate the AC1 statistic using the coefficient agreement formula.

$$AC1 = \frac{P - e(\gamma)}{1 - e(\gamma)} \quad (5)$$

- Based on the interpretation of Gwet's AC1 results in Table I, the results of the AC1 statistical calculations are grouped into the Kappa index, which can be seen in Table II.

TABLE II. GWET'S AC1 AGREEMENT PROPORTION INDEX

Kappa Index	Agreement Proportion
< 0,00	Less than chance-agreement
0,01 – 0,20	Slight agreement
0,21 – 0,40	Fair agreement
0,41 – 0,60	Moderate agreement
0,61 – 0,80	Substantial agreement
0,81 – 1,00	Almost perfect

III. DATASET

The data used in this research consists of two kinds of artifacts. These two artifacts are part of the artifacts formed from the artifacts identification process. In addition, both artifacts are in the form of text documents for simplicity. The Step Performed artifact documents contain words from the Step Performed in the Use Case Description. Meanwhile, the Sequence Diagram artifact documents contain objects from the Sequence Diagram artifacts. The Step Performed artifacts from the Use Case Description and Sequence Diagram of the SRS Digibrary consists of 14 documents for each artifact. Hence, the processed data in this study consists of 28 documents, as shown in Table III and Table IV.

TABLE III. IDENTIFICATION OF STEP PERFORMED ARTIFACTS

No	Step Performed		Artifact ID
	ID	Name of Use Case Description	
1	SP01	Digibrary Registration	d1
2	SP02	Digibrary Login	d2
3	SP03	Searching Book Collection	d3
4	SP04	Borrowing Book Collection	d4
5	SP05	Extend Borrowing Book Collection	d5
6	SP06	Return Book Collection	d6
7	SP07	Input Book	d7
8	SP08	View book	d8
9	SP09	Change book	d9
10	SP10	Delete book	d10
11	SP11	Approve Loan Book Collection	d11
12	SP12	Approve Return Book Collection	d12
13	SP13	Approve Extend Loan Book Collection	d13
14	SP14	Digibrary Logout	d14

TABLE IV. IDENTIFICATION OF SEQUENCE DIAGRAM ARTIFACTS

No	Sequence Diagram		Artifact ID
	ID	Name of Use Case Description	
1	SD01	Digibrary Registration	d15
2	SD02	Digibrary Login	d16
3	SD03	Searching Book Collection	d17
4	SD04	Borrowing Book Collection	d18
5	SD05	Extend Borrowing Book Collection	d19
6	SD06	Return Book Collection	d20
7	SD07	Input Book	d21
8	SD08	View book	d22
9	SD09	Change book	d23
10	SD10	Delete book	d24
11	SD11	Approve Loan Book Collection	d25
12	SD12	Approve Return Book Collection	d26
13	SD13	Approve Extend Loan Book Collection	d27
14	SD14	Digibrary Logout	d28

IV. METHODOLOGY

Based on Fig. 1, there are main activities carried out in this study, namely:

A. Requirements Elicitation

Requirements elicitation activities are to obtain data regarding the requirements needed in building and developing software. Interviews and discussions were conducted with stakeholders regarding requirements that must be provided and implemented during the software development.

B. Artifacts Identification

The results of requirements elicitation will be identified in the SRS documents needed in the research. Identifying artifacts that will be formed from the results of requirements elicitation consists of Use Case Diagrams, Use Case Descriptions, Steps Performed, and Sequence Diagrams. Each artifact identification that will be done has the following explanation:

- Use Case Diagram describes the flow of actor activities using the features in the software until all the features reach their goals.
- Use Case Description explains the activities carried out in the Use Case Diagram.
- Step Performed displays the steps carried out in working on the feature and includes part of the Use Case Description.
- Sequence Diagram displays interactions between objects when working on a feature in the system.

C. Artifacts Extraction Using Text Pre-processing

The extraction process is in the Step Performed and the Sequence Diagram. The results of the extraction of the Step performed and Sequence Diagram are divided into several documents according to the features in the Digibrary application and coded according to the type of each document. Then the artifact extraction results are followed by the text pre-processing process, which is one of the text mining methods.

The text pre-processing process consists of four main processes as follows:

- Case Folding is converting every word in the processed document into lowercase.
- Tokenizing is breaking sentences into words in the documents being processed.
- Stemming removes affixes to each word so that each word turns into a basic word in the processed document.
- Removing Stop Words removes common words that do not provide essential information in the processed document.

D. Document weighting and Consistency Measurement

Before calculating the dataset using the Cosine Similarity formula, the Term Frequency and Inverse Document Frequency processes must be carried out after the text pre-processing activity. Term Frequency is a matrix of the number of word occurrences in the text. Meanwhile, Inverse Document Frequency maps the inverse of document frequency values to the term frequency matrix.

E. Testing and Fixing Artifacts

Testing Artifacts includes validity and reliability tests to prove the feasibility of the obtained conformity results. This testing process uses Gwet's AC1 formula, which consists of two stages: calculating the observed agreement using Gwet's AC1 formula with Python and an expert questionnaire. The questionnaire is a closed question type, so the answers provided are "Yes" or "No".

Fixing Artifacts is after obtaining the results of consistency calculations and testing the validity and reliability between the Step Performed and the Sequence Diagram. Artifacts that were repaired were seen based on the consistent measurement results of the two artifacts in the Use Case, which had the smallest consistency value with repairs following the rules of the Unified Modeling Language (UML) [7, 9, 10, 11, 12].

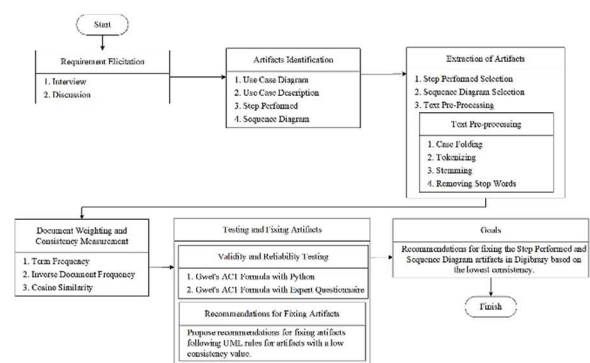


Fig. 1. Research Activities

F. Goals

Goals are obtained after recommendations for improving the Step Performed and Sequence Diagram artifacts were based on the lowest value of consistency. This process shows that the purpose of this research has been achieved.

V. RESULT AND DISCUSSION

A. Requirements Elicitation

The requirements elicitation documents formed from the results of the requirements elicitation process are used as a guide in forming Use Case Diagram artifacts and Use Case Descriptions as the primary artifacts in this study. This document contains conversations between the developer and the client regarding the requirements and specifications of the designed application.

B. Artifacts Identification

1) Use Case Diagram based on Requirements Elicitation

Based on the Requirements Elicitation document, the actors and features in the Digibrary application can be determined. The snippet of Use Case Diagram of Digibrary application can be seen in Fig. 2.

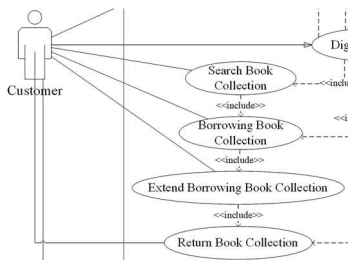


Fig. 2. The Snippet of Use Case Diagram Digibrary

2) Use Case Description based on Use Case Diagram

After the Use Case Diagram is formed, a more detailed explanation of each Use Case in the artifact can be built into the Use Case Description artifact. Snippet 14 Use Case Description Digibrary can be seen in Fig. 3.

Use Case Name	Description	Id
Digibrary Login	This use case gives permission for customer to register their data with the Digibrary application. To access this use case, the customer has opened the home page and does not have an account yet. Then, the customer clicks the login button on the home page. After the customer accesses the registration page, the customer clicks on the words "create an account" and fills in the registration column. Next, the customer fills in their personal data and clicks the submit button to confirm data and send data to Digibrary application. When customer has successfully registered an account, the Digibrary application displays a login page.	UC01
Step Performed	1. Customer opens the home page in Digibrary	

Fig. 3. Use Case Description Digibrary

3) Step Performed based on Use Case Description

Then, Step Performed artifacts are obtained from one part of the Use Case Description for each Use Case of the Digibrary application. One of the 14 Steps Performed Digibrary artifacts can be seen in Fig. 9.

4) Sequence Diagram based on Use Case Description

In this section, Sequence Diagrams are created based on the Use Case Description development results by carrying out design specifications that contain interactions between objects in the system in carrying out a Use Case. One of the 14 Sequence Diagram Digibrary artifacts can be seen in Fig. 10.

C. Artifacts Extraction Using Text Pre-processing

The Step Performed and Sequence Diagram Digibrary artifacts extraction process consists of three main steps. The several steps of the artifacts extraction process are as follows:

1) Selection of Step Performed

Artifact Use Case Description Digibrary is selected only on the Step Performed section as one of the datasets of this research so that sentences with the structure of subject, predicate, and object are produced as datasets for which consistency will be measured. Then, the selection results of Step Performed Digibrary are labeled for each Use Case, the snippet of which can be seen in Fig. 4.

No	Document Labels	Artifacts ID
5	SP05 Customer chooses the collection of books that he or she wants to	D5
7	SP07 Admin clicks on the book data form that he or she wants to input	D7
9	SP09 Admin clicks the edit button on the book he or she wants to change	D9
11	SP11 Admin selects a collection of books on the loan report page. Admin	D11
13	SP13 Admin selects a collection of books on the loan report page. Admin clicks the "Approve" button in the loan report page. The loan report page displays an approval confirmation pop-up. Admin clicks the "Confirm" button on the loan report page. The loan report page displays a collection loan report with the changed return date of the borrowed book.	D13
14	SP14 Admin and customer click "Logout" button in the dashboard page. The dashboard page redirects admin and customer to the	D14

Fig. 4. The Selection Results of Step Performed Digibrary

2) Selection of Sequence Diagram

Meanwhile, Sequence Diagram Digibrary extraction is done by filtering objects from the Sequence Diagram Digibrary in each Use Case. A snippet of the Sequence Diagram Digibrary selection results can be seen in Fig. 5.

No	Document Labels	Artifacts ID
1	SD01 Customer. Home Page.: Login Page: view. Login Page: controller.	D15
6	SD06 Customer. Return page: view. Return Page: controller. Collection	D20
11	SD11 Admin. Loan Report page: view. Loan Report page: controller.	D25
12	SD12 Admin. Return Report page: view. Return Report page: controller.	D26
13	SD13 Admin. Loan Report page: view. Loan Report page: controller.	D27
14	SD14 Admin and Customer. Dashboard Page.: Home Page.: controller.	D28

Fig. 5. The Selection Results of Sequence Diagram Digibrary

3) Text Pre-Processing

Twenty-eight documents were from the labeled dataset, followed by the text pre-processing stage. This stage consists of four main activities: Case Folding, Tokenizing, Stemming, and removing Stop Words that use the Python programming language with the Natural Language Toolkit (NLTK) library. Snippets of the text pre-processing results for this research dataset can be seen in Fig. 6.

```
{'custom': 19, 'open': 44, 'home': 34, 'login': 39, 'button': 9, 'creat': 18, 't': 17, 'column': 15, 'ha': 33, 'provid'}
{'custom': 5, 'home': 7, 'page': 11, 'gistr': 12, 'admin': 0, 'dashboard': 6, 'input': 8, 'book': 1, 'loan': 9, 't'}
```

(a) Step Performed (b) Sequence Diagram

Fig. 6. The Snippet of Text Preprocessing Results for Research Dataset

D. Document Weighting and Consistency Measurement

After text pre-processing on the dataset, the document extracted from Step Performed and the 14 documents extracted from the Sequence Diagram on the SRS Digibrary were measured for consistency through the following steps.

1) Term Frequency

The process of forming the term frequency matrix is before measuring consistency. This matrix will contain the number of occurrences of each word in the processed document text.

2) Inverse Document Frequency

The term frequency matrix obtained will be followed by an inverse document frequency process to map values to the matrix. Therefore, the matrix's value ranges from 0 to 1.

value of the consistency calculation. Requirements elicitation activities and identification of Sequence Diagram artifacts by labeling documents d1 to d14. Then, identification of the Step Performed by labeling d15 to d28 on the SRS Digibrary. The extraction value is generated through text pre-processing as a reference for consistency between the Step Performed and the Sequence Diagram of 0.968165, the highest value found in the SP14 document against SD14. The results of the validation and reliability tests using Gwet's AC1 formula through the Python programming language produced a Kappa Score worth 0.500000. Meantime, the validation and reliability test results using Gwet's AC1 formula through expert questionnaires were worth 0.159998. Based on the test results for the two processes, there is a difference of 0.340002. Based on the artifact consistency calculations being compared, the artifact with the lowest consistency measurement value is 0.840124 in the comparison of document 09 (d9) for Step Performed, which consists of seven steps, with document 23 (d23) for Sequence Diagram, which consists of three objects (Admin, Collection page: view, and Collection page: controller) recommendations for fixing artifacts are made for these artifacts.

For future work, improvements to the Digibrary application SRS can be carried out immediately so that the measurement results for the consistency of the value get closer to 1 so that the SRS content of the Digibrary application is more consistent. Then, the results of improving the SRS Digibrary can be measured for consistency through the Cosine Similarity method or other methods, such as the Overlap Coefficient. Apart from that, the text pre-processing stage of changing affixed words into basic words can be done by methods other than stemming, such as lemmatization.

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